Technical Log

Music Analysis Tool

Overview

Import an audio file and return key signature, time signature, and tempo.

Design Limitations

Feature	Status	Notes
Tempo	Reliable	Less accurate on songs with large tempo shifts
Key	Semi - Reliable	Key Detection System can differ from K-S; typically will include the correct Key within the top 2-3 guesses, if not 1st
Mode	Semi - Reliable	Certain modes are easier to detect, but not all modes are used as a base for songs; the feature is an educated guess
Time Signature	Limited	Overall unreliable, especially for synthesized or melodic audio. May try to implement the future

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Known Issues

- Krumhansl-Schmuckler frequently confuses relative major/minor keys
- Time signature detection is fundamentally difficult without strong rhythmic accents
- Mixolydian and Phrygian are typically identified together, but this is one-directional (Mixolydian scale gets flagged as Phrygian, but Phrygian scales are correctly identified)
- Locrian has issues being identified using a pure scale.
- Lack of songs to test all modes; accuracy using real audio is not currently guaranteed.

Custom Key Detection

Built a custom profile-based key estimator using Pearson correlation. Returns top 3 results with confidence scores—Outperforms music21's built-in Krumhansl-Schmuckler on several test cases, particularly for harmonic minor keys. Currently implements 108 profiles across 9 scale types: Major, Natural Minor, Harmonic Minor, Dorian, Phrygian, Lydian, Mixolydian, Phrygian Dominant, and Locrian (12 keys each).

Future extensions: Chord progression generation, Chord progression identification, Time signature identification

Dev Log

Development Log

5/29/26 — Initial Setup & Core Features

Setup

- Created `test.py` to verify library imports and versions
- Resolved Python interpreter mismatch between terminal and VS Code

Features Implemented

- Tempo detection via `librosa.beat.beat_track` + interval method as backup
- Smoothed tempo curve over time using `scipy.ndimage.uniform_filter1d`
- Chromagram visualization — heatmap and bar chart
- Key detection via Krumhansl-Schmuckler (music21)
- Custom key detection system — 36 profiles across major, natural minor, and harmonic minor

5/30/26 — Fixing Errors and Extending Features

Errors

- Resolved Logic error within the base c_natural_minor that gave weight to A instead of G#
- Updated logic for custom key identities: 1st, 3rd, and 5th notes are given 6.0 weighting, other notes in the mode are given 3.0, and the rest are given 1.0

Features Implemented

- Added new sample scales to test differences between librosa detection and my custom-built key detection
- Dorian, Phrygian, Locrian, Lydian, Mixolydian, and Phrygian Dominant were implemented into custom key detection
- Adjusted Locrian profile — diminished 5th (F#), weight reduced from 6.0 to 3.0 to reflect the instability of the interval. Correctly identifies C Locrian after adjustment. However, this adjustment then identifies C Dorian as C Locrian. Reverted to 6.0
- Updated Key Signatures to use the standard naming, although some naming for modes may be incorrect.
- Designed a Parent Key identification to pair with mode identification

Current Limitations

- Mixolydian/Phrygian Ambiguity – Modes with the same parent scale (e.g., C Mixolydian and A Phrygian) produce nearly identical scores when tested using a scale
- Locrian – Unreliable detection due to diminished 5th
- **General note** — Chromagram-based key detection fundamentally cannot detect tonal center, only pitch content.
- Lack of proper songs to test all modes.

5/31/26 — Testing and debugging Features

Errors

- The parent key and certain modes give ambiguous results; the weighting of notes may need to be redesigned.
- Mode identification using the parent key has inadvertently become a modal alignment & coherence check, compared to the original design.

Features Implemented

- Changed tonic weight to 10.0; Most music will feature the tonic the most heavily, hence this change
  - Led to improved recognition for the C major chord progression
  - Increasing tonic weight overall seems to improve for songs that have a strong tonic presence, but it is not good for more ambiguous songs like Madonna's Don't Tell Me
  - May need to find an algorithmic way to optimize weights
- Implemented a new confidence system, returning different types of results depending on confidence.
  - Good for songs like Don't Tell Me
- Fixed Coherence Check to have standardized scale identities (e.g., Bb instead of A#)
- Tested tonic weight at 8.0, notes not in the scales changed from 1.0 to 0.2 weight.
- Standardized chords created using `https://www.onemotion.com/chord-player/` and `hooktheory.com`

Current Limitations

- Progressions that lead with and emphasize the tonic and characteristic chord of the mode are correctly identified. Progressions that bury the tonic among complex harmony get misidentified.
  - Example: The Two F# Phrygian Chord Progressions included led to one being identified correctly, the other as a harmonic minor.

- Custom chord progressions may feature my personal error in creating progressions that are not actually from the mode; standardized progressions/songs are better for testing
- May need to reframe key/mode detection: Not as a way of detecting for sure, but as the most common/probable framing of the song.

To-Do

- Try moving notes not in mode from 1.0 to 0 - 0.5; test for improved accuracy
- Try implementing a tie-break filter for modes and key
    - Base Check: Tonic and Dominant note energy
    - Character Bonus: Energy of notes unique to the mode (e.g., C in D Mixolydian)
    - Forbidden Penalty: Energy of note that should not exist (e.g., C# in D Mixolydian)
- Potential pivot away from identification:
    - Focus more on creation, not identification, if this becomes too strong an issue

Test Results

5/29/26 - Initial Tests (Major, Minor, Harmonic Only)

Song	Known Key	Known BPM	Custom BPM	Custom Key	K-S Result	Correct?
老張 - No Party for Cao Dong	E Major	135	135.31 - 135.99	E Major	G# Minor Alt: B Major	✅
Tank! - The Seatbelts	C Minor	137-140	135.99 - 137.52	C Minor Alt: C Harmonic Minor	C Major Alt: C Minor	✅

Notes:

- Custom system outperforms K-S on both tests
- Tank! - The Seatbelts
    - Key is ambiguous between sources (Spotify vs Wikipedia)
    - Lower confidence scores overall reflect the harmonic complexity of the jazz arrangement.
    - Tempos are slightly off from 140.
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- 老張 - No Party For Cao Dong
    - Custom system correctly identified E major with high confidence (0.846).
    - Krumhansl-Schmuckler returned G# minor, the relative minor of E major.

5/30/26 - Secondary Tests (Inclusion of Other Modes)

Song	Known Key	Known BPM	Custom BPM	Custom Key	K-S Result	Correct?
Love Me Again - John Newman	G Dorian	127	126 - 126.42	G Dorian Alt: G Minor	G Minor Alt: G Major	✅
Can You Feel the Love Tonight - Elton John	B♭ Major	126	129.2 - 127.36	B♭ Major	B♭ Major	✅
Don’t Tell Me - Madonna	D Mixolydian	100	99.38 - 99.87	D Lydian Alt: D Mixolydian, D Major	D Major	❌

Notes:

- There does not seem to be any issues identifying Dorian, Major, Minor, Harmonic Minor.
- Some of the keys need to be updated to convention: i.e., B♭ Major instead of A# Major (Which is what is currently being used)
- Don't Tell Me
    - D Lydian, D Mixolydian, and D Major are returned. This is a bit confusing, as Lydian and Mixolydian do not really have overlap.
    - Could be the Chromagram identifying A as G# incorrectly, but I'm doubtful based on the heatmap.
- Overall, I need more songs with different keys to test accuracy.

5/31/26 - C major Chord Progression Initial Results

Song	Known Key	Known BPM	Custom BPM	Custom Key	K-S Result	Correct?
C major progression [See Custom Folder]	C Major	90	68 - 71	F Lydian A Minor	C Major	❌

Notes:

- Identification before any major changes to weights
- BPM is strangely off, although this could be a one-off occurrence for chords when isolated
- Notably, F Lydian and A minor use the same notes as C Major; this implies a need to reweigh tonic note values, etc.
- Overall, the custom key detection system also correctly identified C, secondary being F
- Custom chord progressions may have error from me

5/31/26 - Standardized Chords [To the Best of my Ability]

Song	Known Key	Known BPM	Custom BPM	Custom Key	K-S Result	Correct?
A H-Minor Chords	N/A	N/A	N/A	A Harmonic Minor	A Minor	✅
A Minor Chords	N/A	N/A	N/A	A Minor	A Minor	✅
B Locrian Chords	N/A	N/A	N/A	C Major: Alt: B Locrian, G Mixolydian	G Major	❌
C Major Chords	N/A	N/A	N/A	C Major	C Major	✅
D Dorian Chords	N/A	N/A	N/A	A Minor Alt: A H-Minor, A Dorian	A Minor	❌
E Mixolydian Chords	N/A	N/A	N/A	E Mixolydian	E Major	✅
E Phrygian Dominant Chords	N/A	N/A	N/A	F Lydian Alt: A H-Minor A Minor	A Minor	❌
G Phrygian Chords	N/A	N/A	N/A	G Phrygian	G Minor	✅
F Lydian	N/A	N/A	N/A	C Major Alt: C Lydian C Mixolydian	C Major	❌

Notes

- Used 8.0 weights for the Ionic during testing.
- Minor, H-Minor, and Major rarely have issues being identified.
- B Locrain:
    - Surprisingly did show up as a close second in the Alt; Identification of this mode is not a priority due to how rare it is.
- D Dorian:

    - Given the top 3 were all A, the progression itself likely doesn't stress D as the tonic. This does not seem to be an issue for analyzing Dorian songs like Love Me Again however.
    - Either this progression was bad, or Dorian songs have more information pointing towards the tonic outside the progression.
    - Testing with just the first half of the progression still yields F Lydian as the first result by a wide margin.
- E Phrygian Dominant Chords:
    - Similar Issues to D Dorian; Could be a progression issue on my part
- F Lydian
    - This one is for sure a chord progression issue: The Chords themselves truly seem to resolve to C.
- Main Takeaway:
    - My use of Key Detection for modes may be inherently flawed. It may be better to use the key approach as an approximate guess of the mode, which may possibly be inaccurate.